

See why the series in the definition of algorithmic probability always converges

Convergence of the series defining algorithmic probability

The convergence of $P(s) = \sum_{p: U(p)=s} 2^{-|p|}$ can be proven using [Kraft's inequality](#).

In a nutshell:

Let's first observe that there is a one-to-one correspondence between a finite binary string p and an interval $[0.p, 0.p + 2^{-|p|}]$, where $|p|$ is the length of p and $0.p$ designates a real number by its binary development.

The length of such an interval is $2^{-|p|}$.

If you think about it, you'll see that all these intervals are disjoint, because all programs p are considered to be prefix.

As a consequence, the sum is necessarily smaller than 1.

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